



EDUCATION

BTech - Robotics & Autonomous Systems	Plaksha University, Mohali	8.68/10	2027
12th - CBSE	DAV Public School, Haldia	94.4%	2022
10th - CBSE	DAV Public School, Haldia	94.6%	2020

EXPERIENCE

- AI Software Engineering Intern | IndiaMARTJune 2026 - July 2026
 - Built an **automated pipeline** to estimate standard product prices at scale, matching listings across multiple internal and external sources using rule-based and LLM-driven techniques.
 - Migrated key data sources to cloud storage (**GCP**), improving price coverage through data integration and brand-aware filtering.
 - Improved match accuracy from ~60% to **83%** through refined matching logic and data source integration.
- Summer Intern | Robotics Lab, PlakshaJune 2025 - Aug 2025
 - Developing an **autonomous robot** for crop inspection, utilising sensors and **Machine Learning** for leaf-level disease and pest detection in commercial and horticultural crops.
 - Collaborated on optimising navigation, data analysis, and real-time crop health monitoring to advance **precision farming**.
- Senior Teaching Assistant | Young Technology Scholar, PlakshaMar 2025 - Jun 2025
 - Fabricated a **strain sensor** from scratch using conductive materials and other chemicals.
 - Built multiple mini-projects to explore sensor applications, and compared **strain vs flex sensors** for performance and responsiveness.
 - Integrated sensor with a robotic arm to demonstrate **prosthetic control**.
- Summer Intern | Dixon IoT Lab, PlakshaJun 2024 - Nov 2024
 - Working on advancing solutions for **smart traffic and agricultural management systems** under the mentorship of Prof. Srinivasan.
 - Worked with **IoT hardware** and different networking protocols like **MQTT, CoAP, AMQP**.

PROJECTS

- AI Website Assistant for Plaksha University | Prof. SobtiJan2026 - June2026
 - Built a **semantic AI chatbot** for Plaksha University's website with deployment planned for the live website.
 - Implemented a **3-level RAG architecture**: FAQ JSON retrieval, local hybrid retrieval using Chroma + BM25, and Tavily official-domain fallback for weak-context queries.
 - Designed the pipeline for dynamic scraping, content cleaning, reranking, and cost-efficient LLM usage with **open-source embeddings**.
- MediRep | SelfFeb 2026 - March 2026
 - Built a **full-stack AI assistant** for doctors with real-time chat, voice input, and structured clinical outputs.
 - Designed a production-grade **RAG system** integrating drug, interaction, and reimbursement data.
 - Engineered an end-to-end RAG pipeline with custom **guardrails** and **vector database** architecture for safe, reliable responses.
- Intrusion Detection System | Prof. SobtiAug 2025 - Dec 2025
 - Designed and implemented a **multiclass IDS** leveraging **Deep Learning** on tabular network traffic data.
 - Explored various architectures, including **LSTM, CNN, Autoencoders, Knowledge Distillation and Transformer**, achieving best performance with a hybrid **Encoder-Decoder → X-LSTM → CNN** pipeline for spatial-temporal pattern learning.
 - Reached **98.1%** accuracy and F1-score with the final model.
- Crop Health Analysis | Prof. SiddharthFeb 2025 - May 2025
 - Developed a **Deep Learning** model for **multi-crop health classification** using satellite imagery and weather data.
 - Applied **CNN** for spatial-spectral feature extraction and **LSTM** for time-series analysis, comparing ML and DL models on multi-spectral data performance.
 - Reached **80%** accuracy and F1-score with the final model.

POSITIONS OF RESPONSIBILITY

- Class Representative, RAS | Plaksha University
 - Liaised between students and faculty, addressing concerns and ensuring effective communication.
- Environmental Captain | DAV Public School
 - Led green initiatives and environmental campaigns within school.

SKILLS

- Programming Languages:** C++, C, Python, MATLAB
- Web Development:** HTML, CSS, JS
- IoT:** Raspberry Pi, Networking Protocols, ESP32
- Technologies:** Machine Learning, Deep Learning, Gen AI, RAG, MySQL, MCP, GCP, n8n, Reinforcement Learning
- Framework:** PyTorch, LangChain, LangGraph, FastAPI

ACHIEVEMENTS

- Runners-up, Unstop Hackathon** – Developed a soil health monitoring system using IoT and ML for real-time agricultural insights (received incubation opportunity from IIT Ropar).
- Co-founder, UnDeny** – Built an AI-powered platform to help families claim denied medical insurance; ranked Top 4 out of 25 teams.
- Co-author, ASEE Research Publication** – Published a research paper on project-based learning and experiential learning frameworks.
- Certificate of Distinction** – Australian National Chemistry Quiz, Royal Australian Chemical Institute (International Event).